Pulseq News

(basic format and matlab toolbox updates)

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What's new in *Pulseq*

(Matlab Pulseq)

- Requested block duration in addBlock
- max RF amplitude as system property / RF amplitude warning
- (relative) SAR calculation
- Moments and b-factor calculation
- New display options: paperPlot(), plot('stacked'), plotting of triggers
- Plotting & debugging functions for labels & counters
- Mechanical resonances analysis
- Integrated signal simulation (MR0, Koma.jl)

New Features in Format v1.5.0 / v1.5.1

- More information about **RF shapes**
  - RF center stored in the file – more precise sequence trimming
  - Intended RF use stored in the file – reliable k-space calculation for loaded Pulseq files
- More control over **arbitrary gradient** shapes
  - First/last gradient sample stored in the Pulseq file
  - Storing of gradient shapes on time raster oversampled by the factor of 2
- **FOV positioning** functions (Matlab, Python code submitted)
- “Soft delays” extension
  - Allow to retrospectively changing TE, TI, TR, **cardiac delays**, etc…
- **RF shimming** extension
- Rotation extension
  - Allows to specify **3D rotation per block**

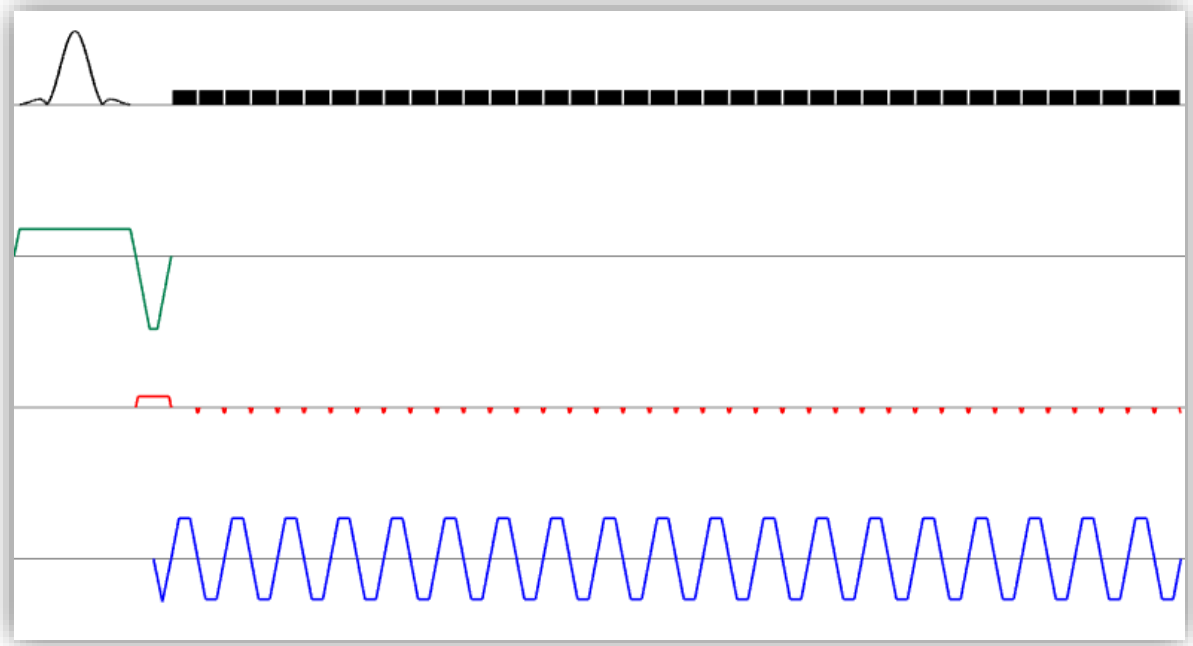
Breaking Format Changes: v1.5.x

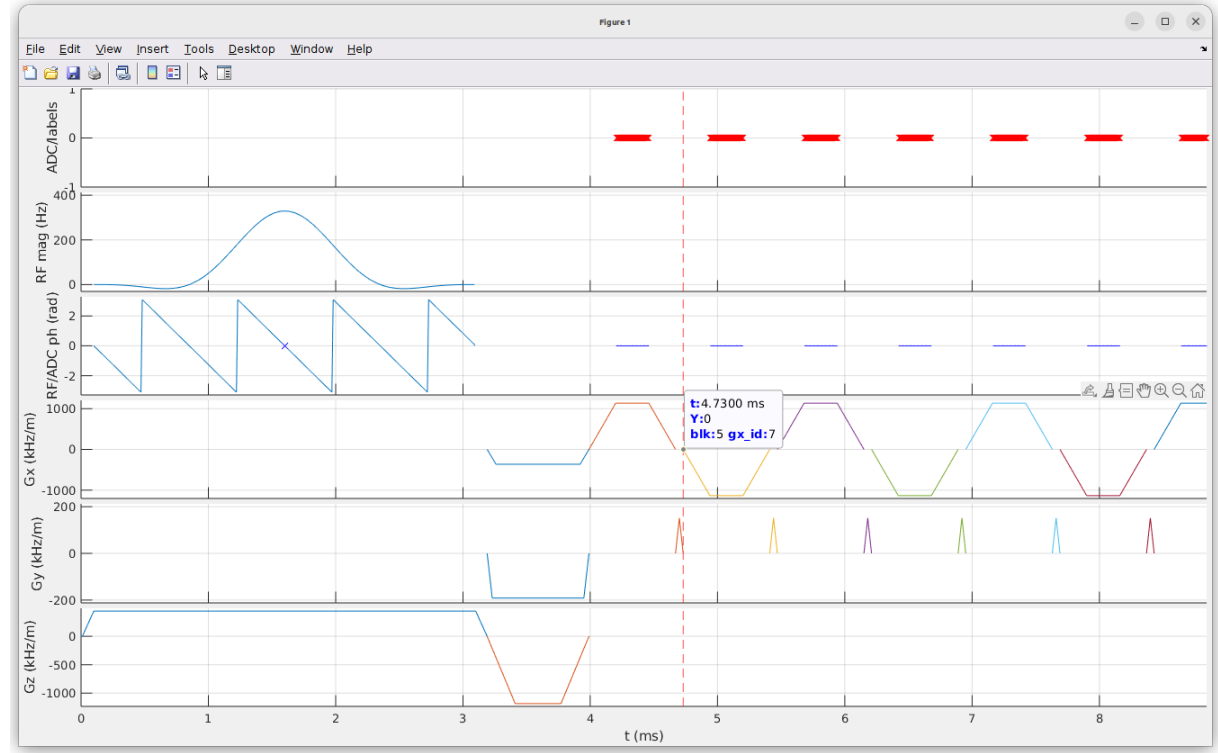
- The file written by seq.write() is automatically in the **new** format
- All **previous formats** can be read by seq.read()
  - Memory storage corresponds to v1.5.x
  - Seq.read() followed by seq.write() will convert the file to the current format
- For **compatibility** with older interpreters use seq.write_v141();
  - If no new features are used the file *should* be 100% compatible with v1.4.1
  - If new format features and extensions are used in the sequence, no 100% compatibility can be guaranteed, e.g. the interpreter *should* be able to ignore unknown extensions
- RequiredExtensions: new concept in v1.5.1
  - Interpreters **must deny** Pulseq files if they do not support these extensions

New Features in Pictures

Plotting Options in Matlab

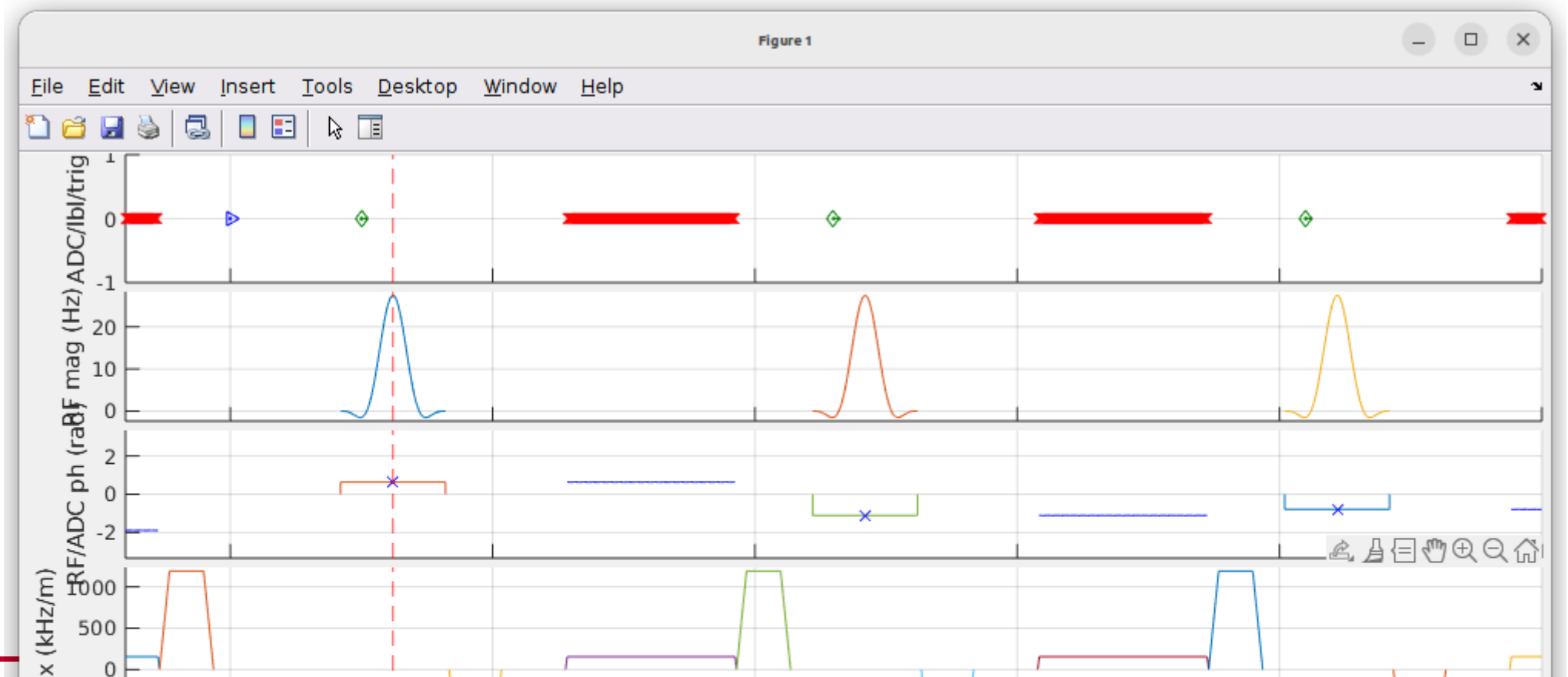
- seq.plot('stacked',1,…);
  - Hairline and DataTip
- seq.paperPlot()



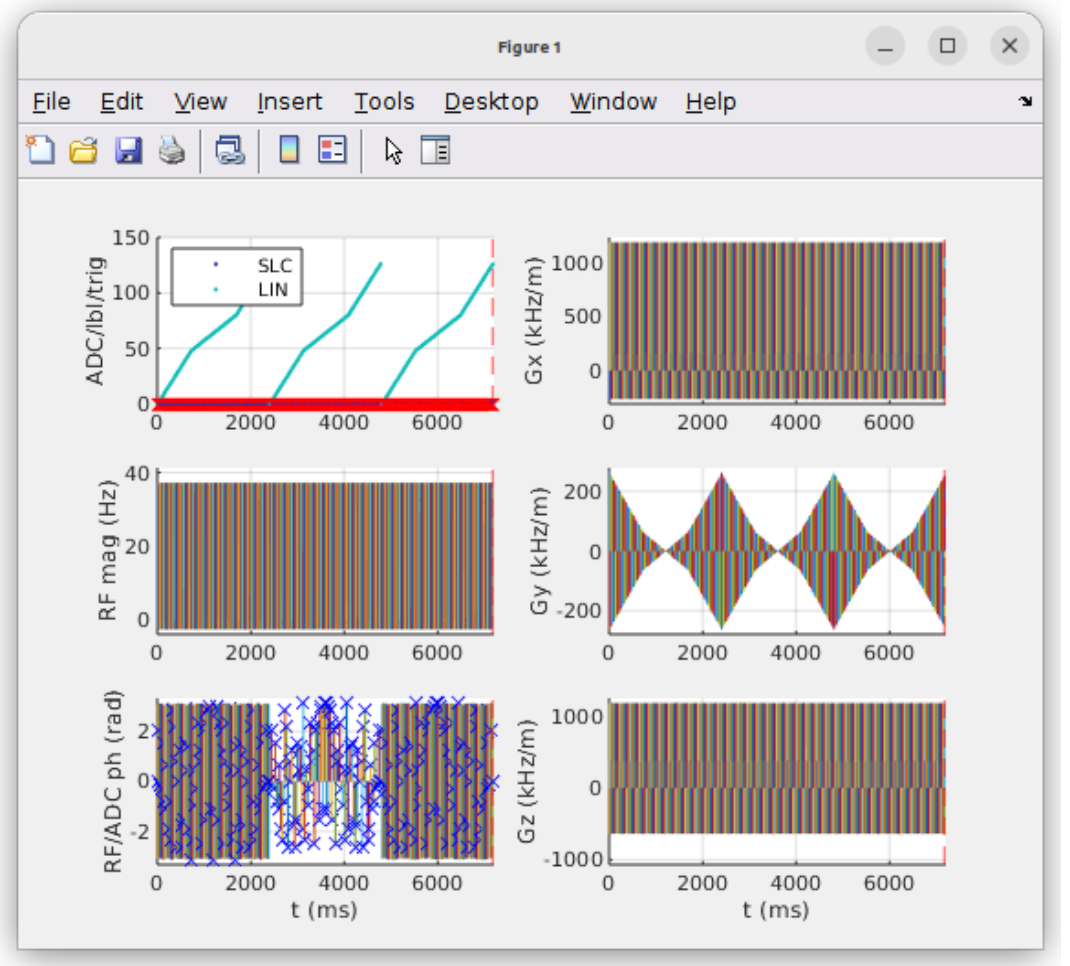


Trigger Display in seq.plot()

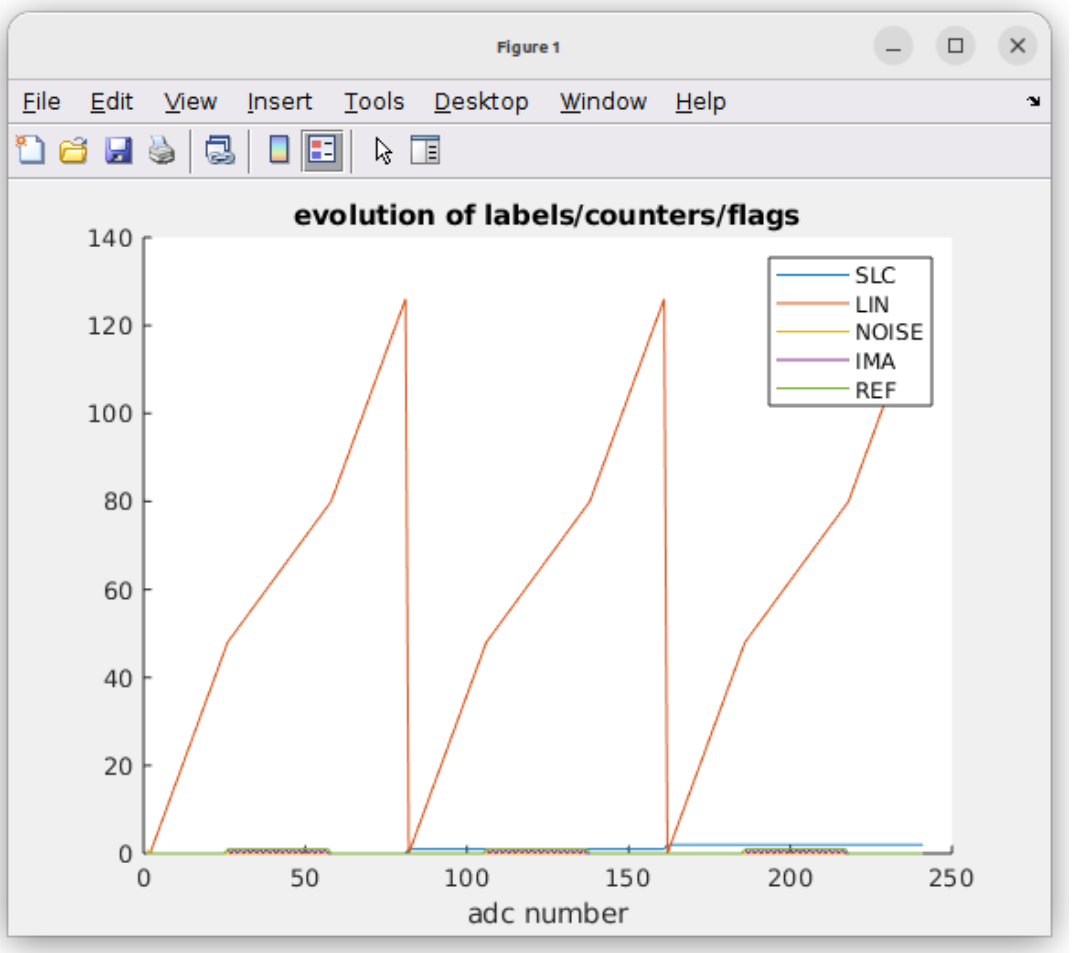
- Triangle: input trigger (e.g. for cardiac gating)
- Diamond: digital output pulse (dash – duration of the trigger event)



Visualizing and debugging label evolution



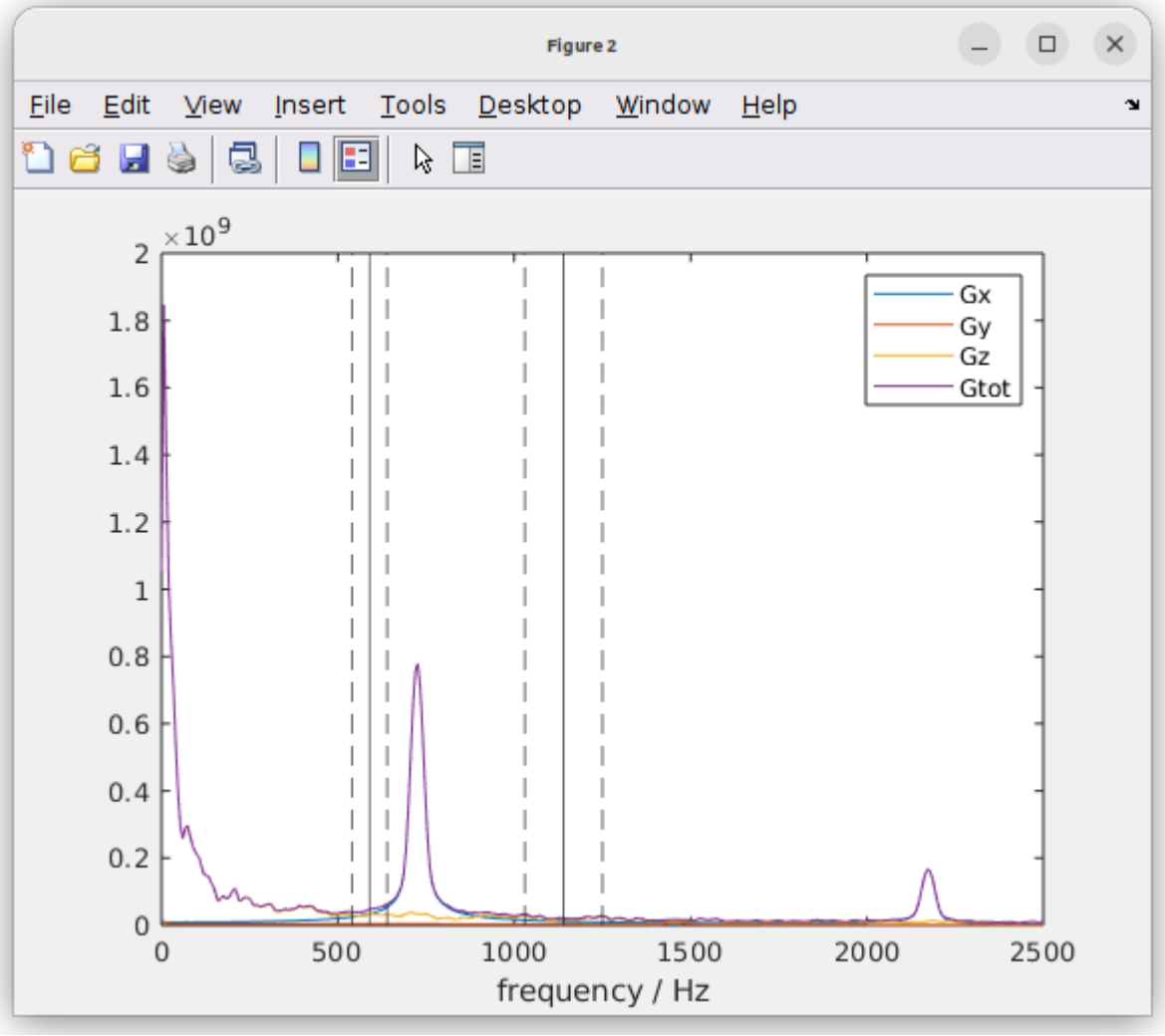
seq.plot('Label', 'LIN,SLC‘,…)



seq.evalLabels(…)

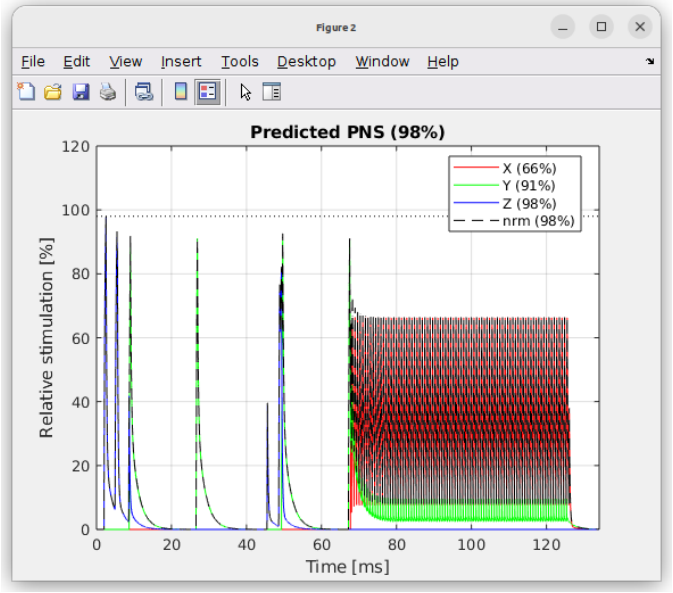
Mechanical resonance analysis

- Calculate gradient spectrogram **`seq.gradSpectrum(…)`**
  - Not anymore in demoUnsorted/gradSpectrum.m
- Possibility to import Siemens scanner properties (here showing Prisma's forbidden frequencies)
- Custom resonance information

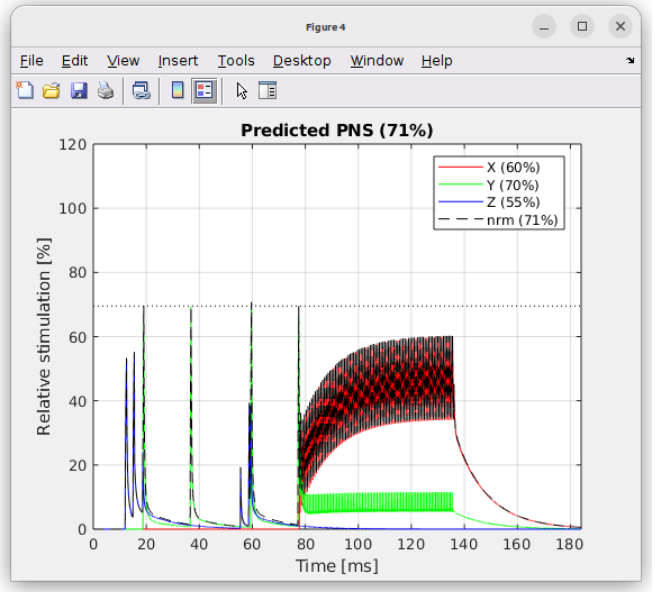


Peripheral nerve stimulation

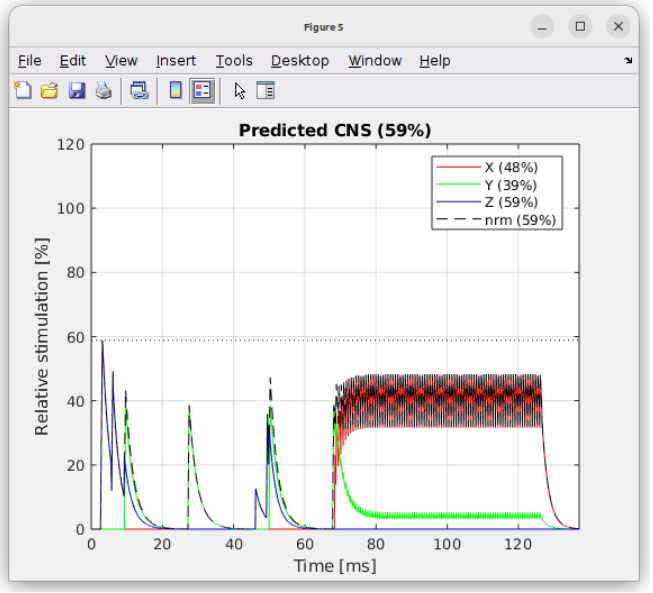
- PNS prediction based on the Siemens' SAFE model
  - Relies on "safe_pns_prediction" by Szczepankiewicz and Witzel see http://github.com/filip-szczepankiewicz/safe_pns_prediction
- Allows to import scanner parameters, 99% similar to IDEA
  - Now support both PNS and CNS (for scanners that have such models)



PNS, Prisma



PNS, CimaX

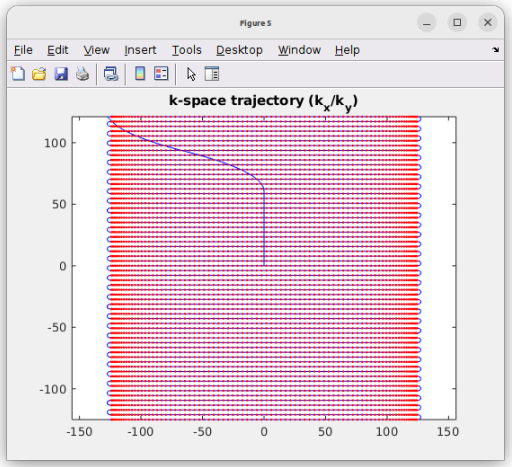


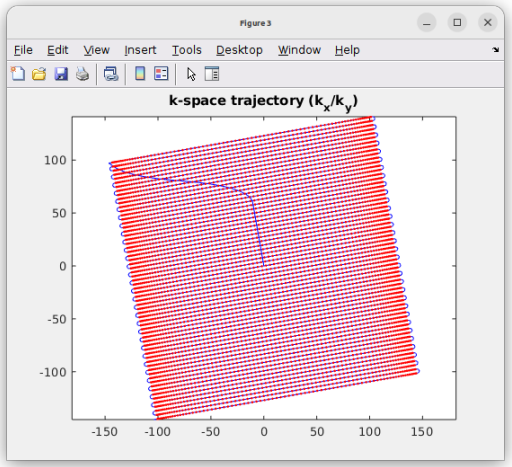
CNS, CimaX

Offline FOV Positioning

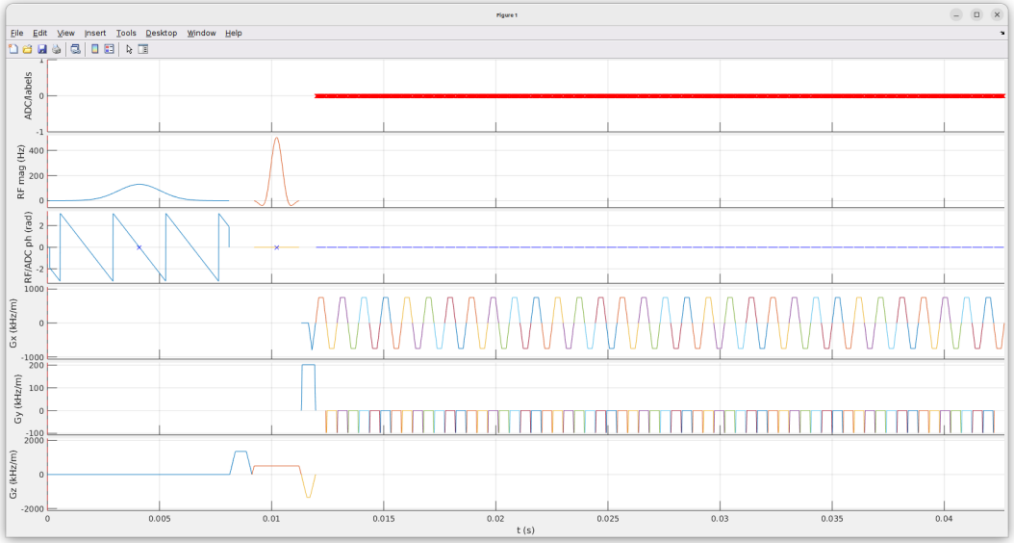
- Really simple to use

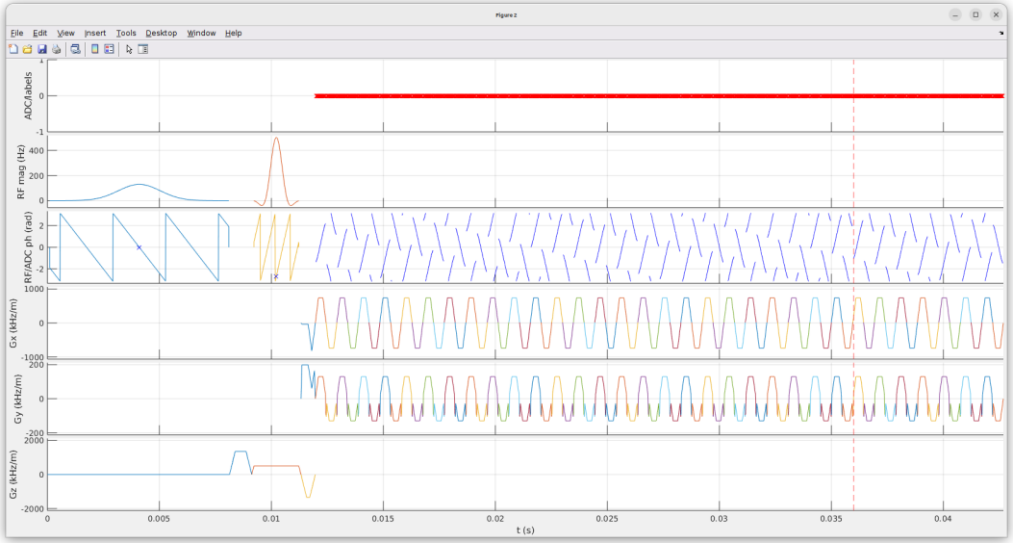
```
>> TF=mr.TransformFOV('translation',...
[0.001,0.01,0.003],'rotation',rotz(10));
>> seq2=TF.applyToSeq(seq);
>> seq2.plot(...);
```





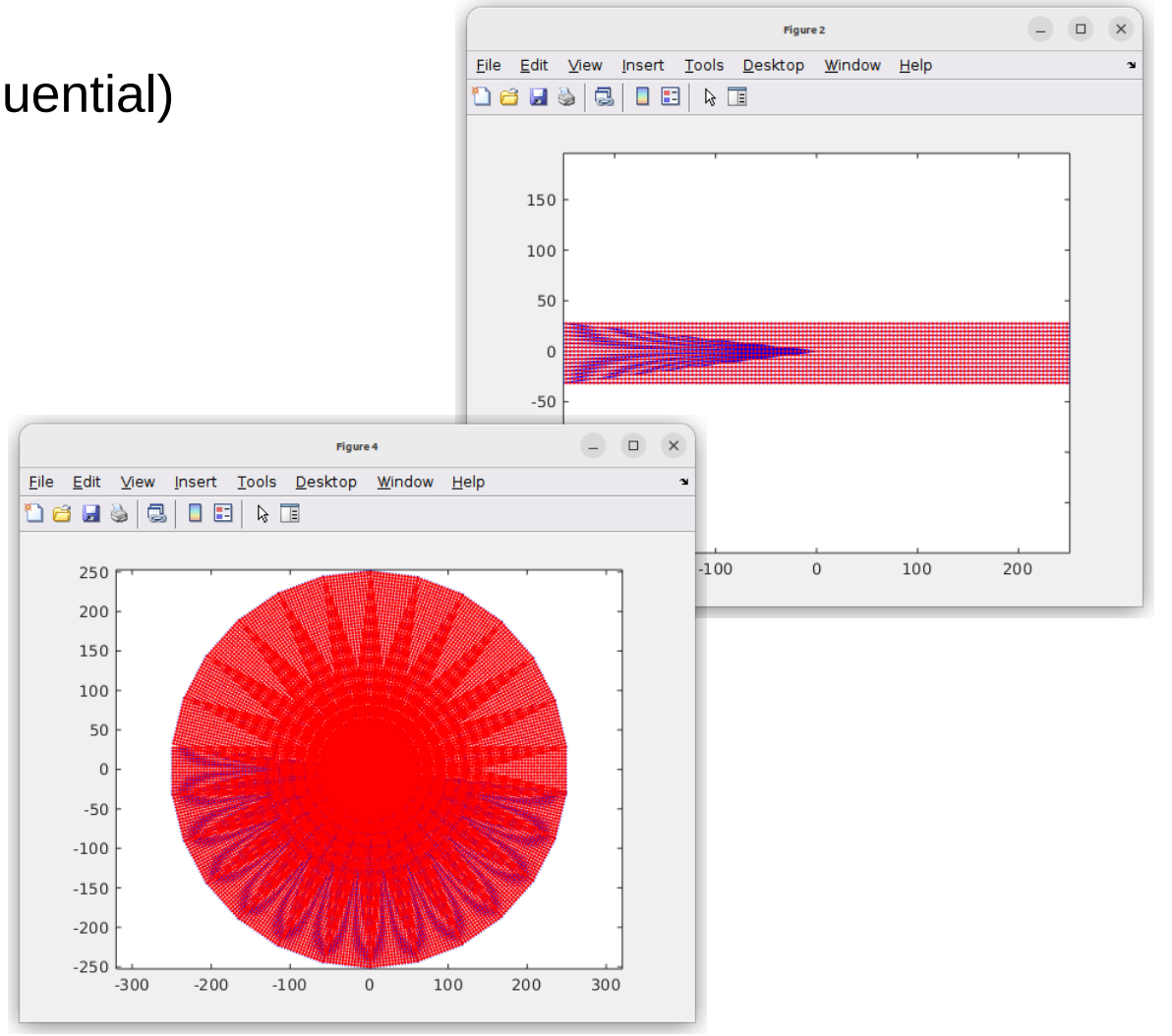
- Applicable to the entire sequence, parts of the sequence, or a block





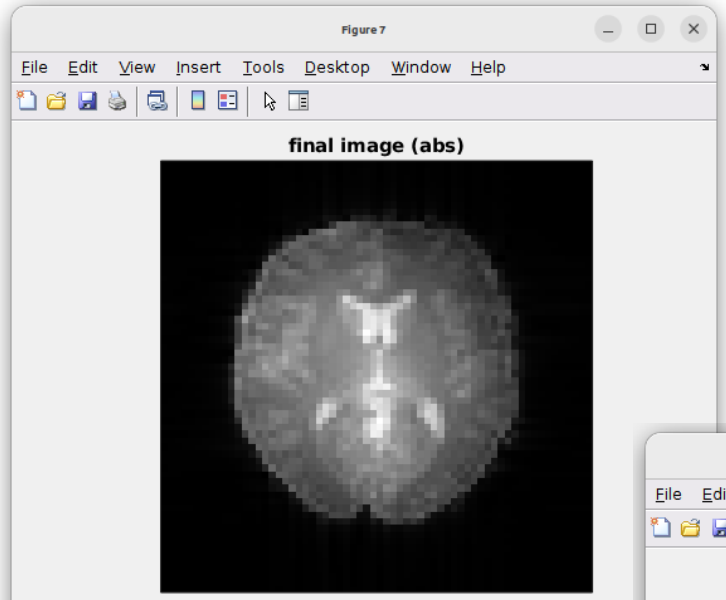
Use FOV tools to simplify sequence design

- Program a single slice
  - **Clone with shift** to add more slices (sequential)
  - See demoSeq/writeEpiRS_label_trans.m
- Program one blade
  - **Clone with rotation** to add more blades
  - See demoUnsorted/mini_gre_prop.m demoSeq/writeTSEprop.m
- See demoSeq/writeSpiral.m for **interleaved spirals**

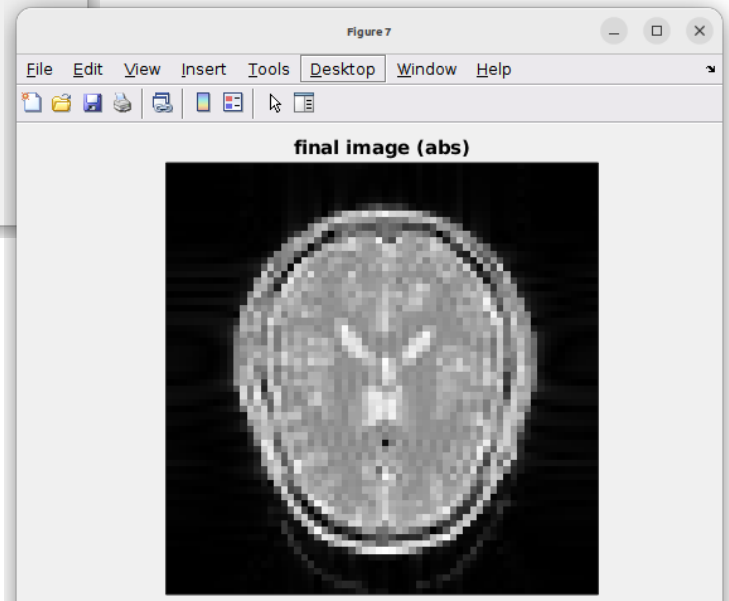


Integrated Data Simulation Tools

- Two prototype functions
  - ‘demoUnsorted/simMR0.m’
  - ‘demoUnsorted/simKoma.m’
- Need standalone installations
  - Python / MR0
  - Julia / KomaMRI
- Simulation time for of 2D sequences between 3..30 seconds



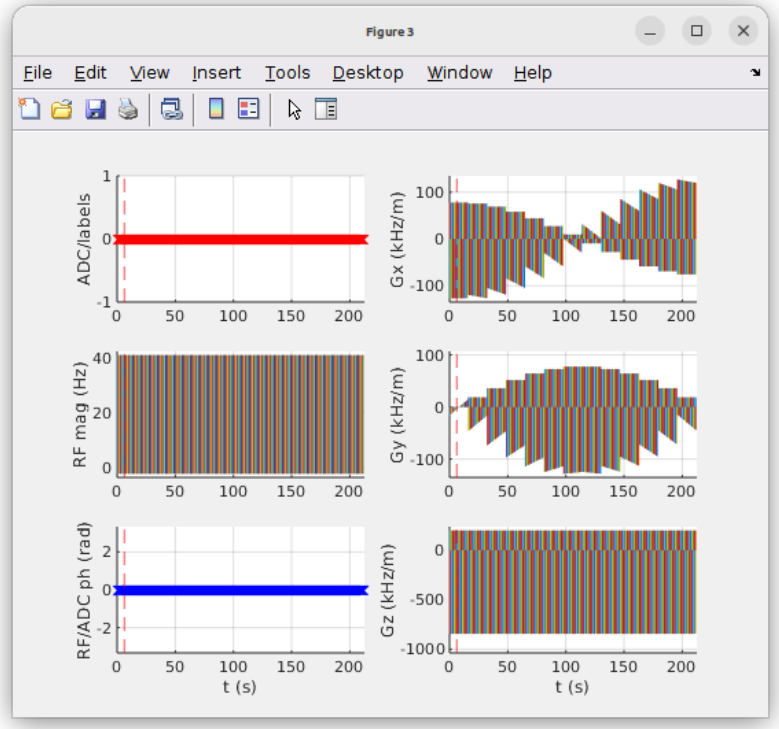
EPI simulation with MR0

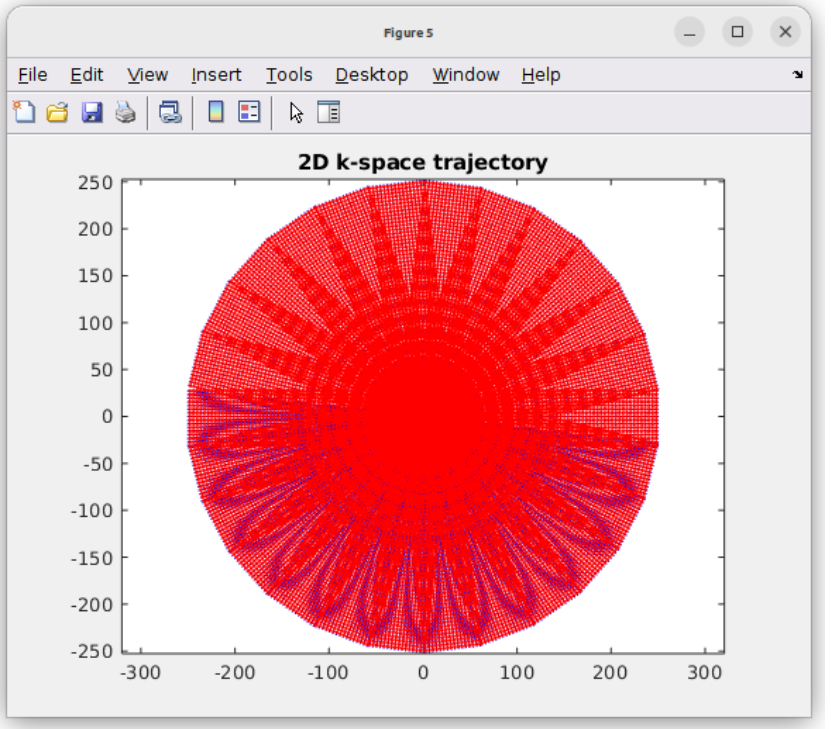


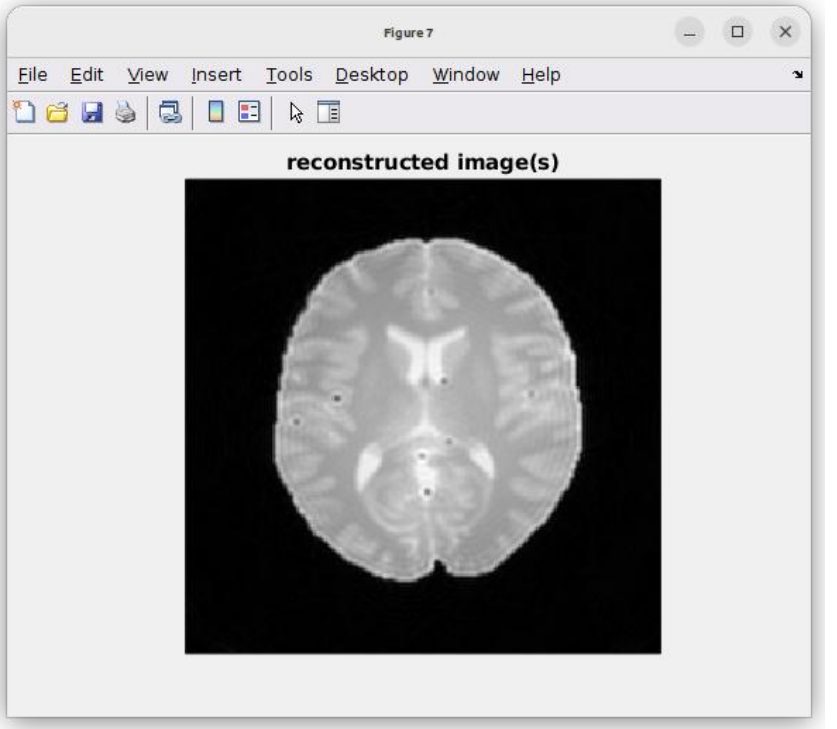
EPI simulation with Koma

Simulation Example: 2D Propeller

- Sequence trivially generated using the FOV positioning functionality
- Simulated in few seconds with MR0
- Reconstructed using standard demo scripts (reconExample2DGD)

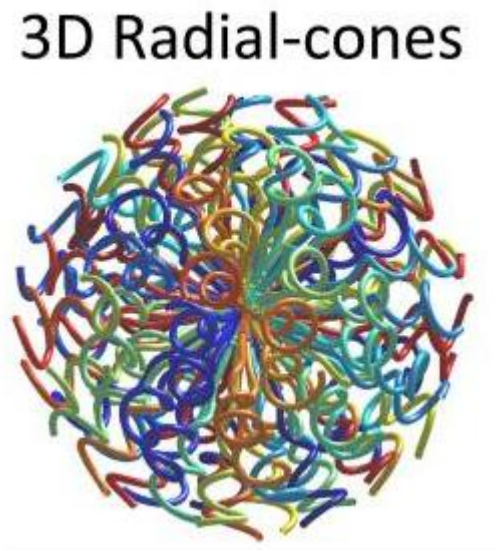


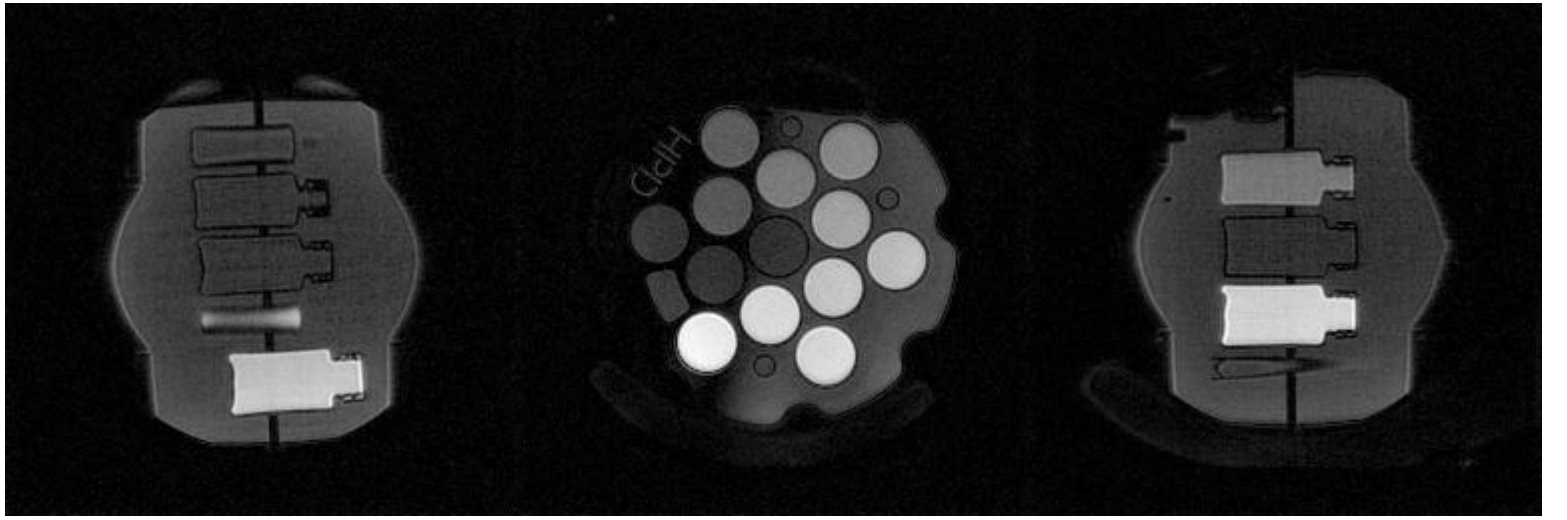




Rotation Extension

- Possibility to add rotation maxtrix/quaternion per block
- Tremendous sequence complexity reduction for 3D multi-shot spirals and alike*





30000 radial-cone arms, TR 4ms, TE 100us, ~2 min acq time
Courtesy of Scott Hoge

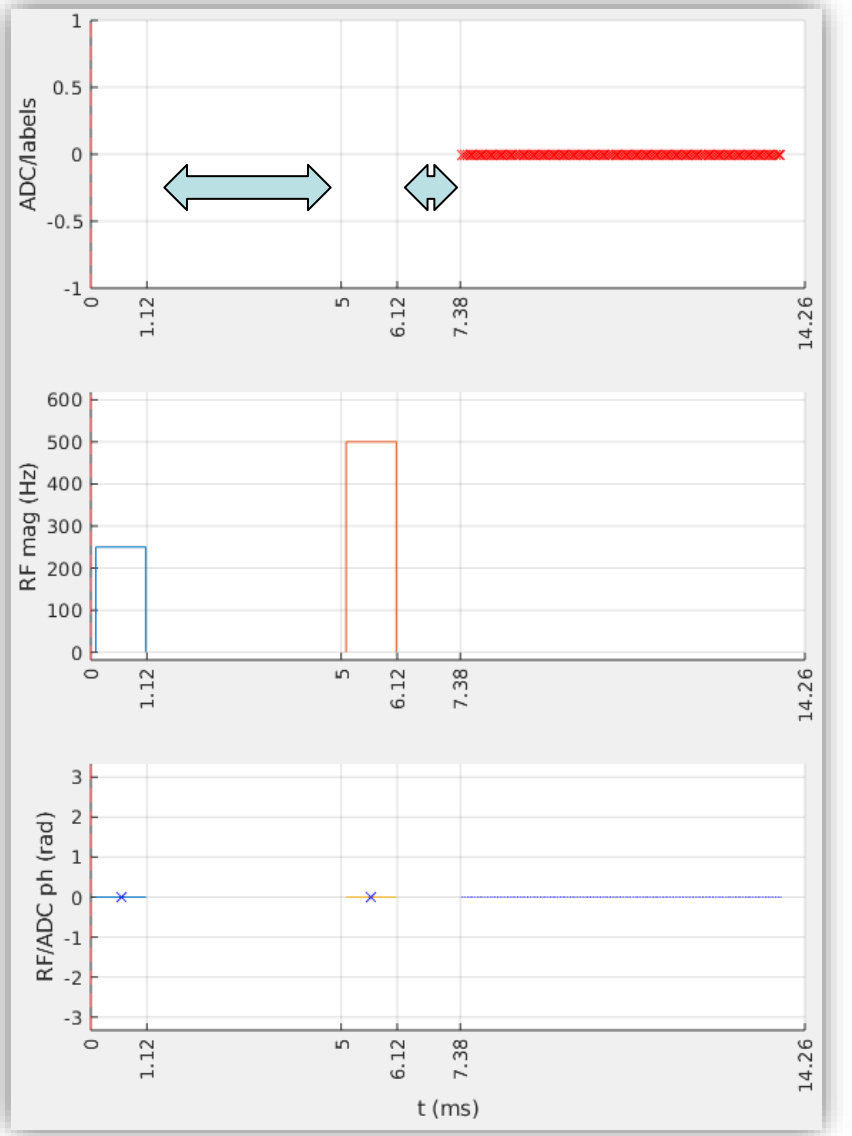
*Johnson KM. Hybrid radial-cones trajectory for accelerated MRI. MRM 2016. doi:10.1002/mrm.26188

Soft Delay Extension

- Any **pure delay block**
  - Empty block containing no gradient or RF events
  - Non-zero duration
- Soft delay extension object: modify block's duration
  - **Rewrite block duration** based on the user input
  - Duration of 0 is allowed
  - Debug/test using seq.applySoftDelay()
- Soft delay is identified by the provided numeric ID
  - A text hint is expected to be provided (same hint for the same numeric IDs)
  - Additional features: offset and factor

Soft Delay Use Cases: Offset & Factor

- Toy example 1: spin echo TE
  - The user provides the desired TE
  - Duration Block_2:
    TE/2 – RFex_dur/2 – RFref_dur/2
  - Duration Block_4:
    TE/2 – RFref_dur/2 – ADC_dur/2
- Toy example 2: TR in multislice sequences
  - Factor = N_slc
  - Offset = actual duration per slice
- Toy example 3: TE & TR adjustment
  - To resolve interactions between multiple soft delays the inner loop needs to have a ‘negative’ filler delay at the end of TR period
- **Main use case: subject-specific cardiac delay**



Implementation Details in Demo Sequences

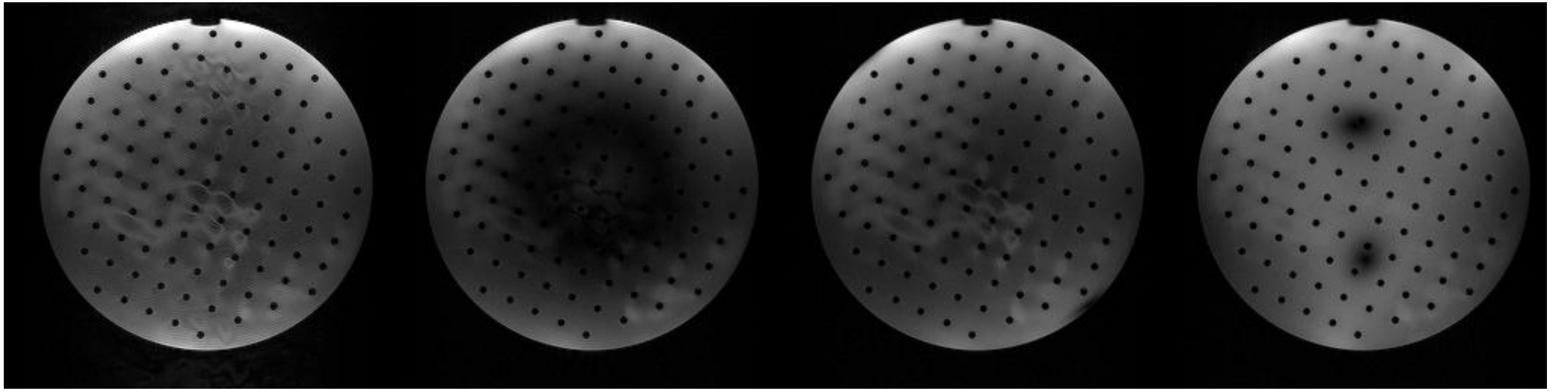
- Matlab Pulseq:
  - demoSeq/writeEpiRS_label_softdelay.m
  - demoSeq/writeSpinEcho_softdelay.m
- Siemens Interpreter
- PyPulseq: WIP

RF Shimming Extension

- *Inspired by the work of Martin Freudensprung & Moritz Zaiss (ESMRMB 2023)*
- Specify channel-vise RF scaling as a complex number
- RF shim is set per block (can be vary throughout the sequence)

```
>> rfs = mr.makeRfShim([1, exp(1i*2*pi/(Nrep)*(r-1))]); % two-channel shim setting
>> seq.addBlock(rf,gz,rfs); % 'rfs' adds RF shim setting to the current block
```

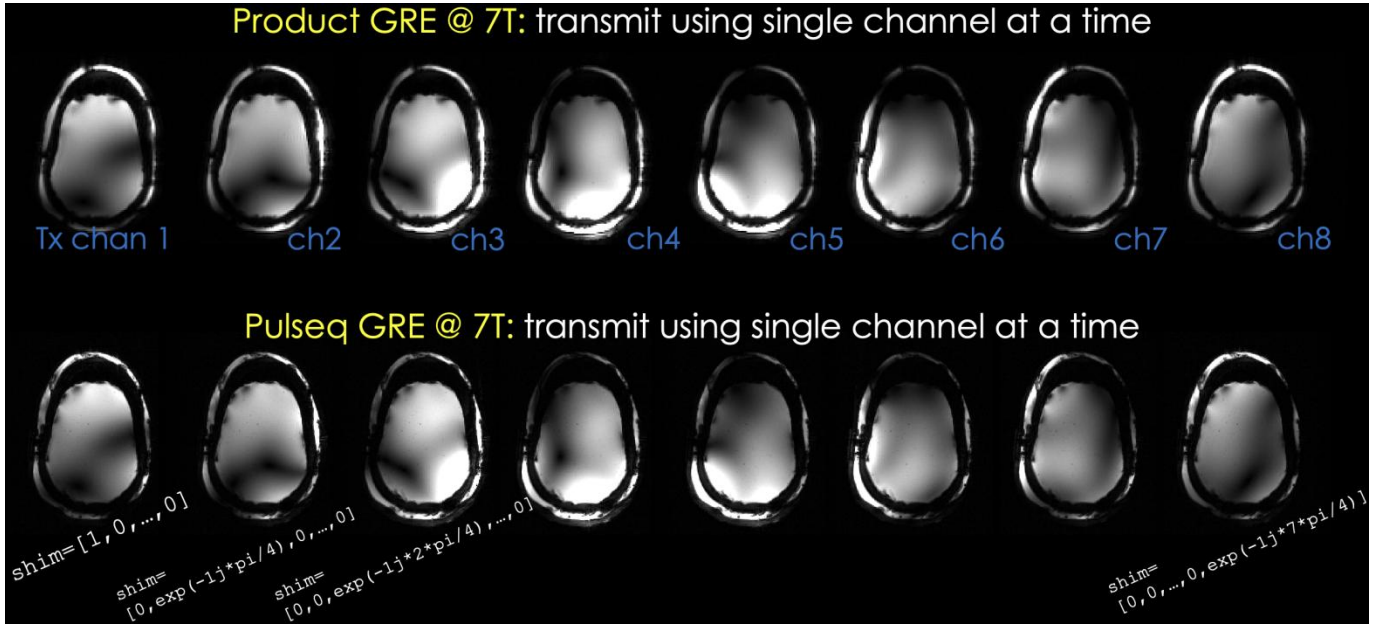
- Tested on 3T 2-channel pTx scanner:



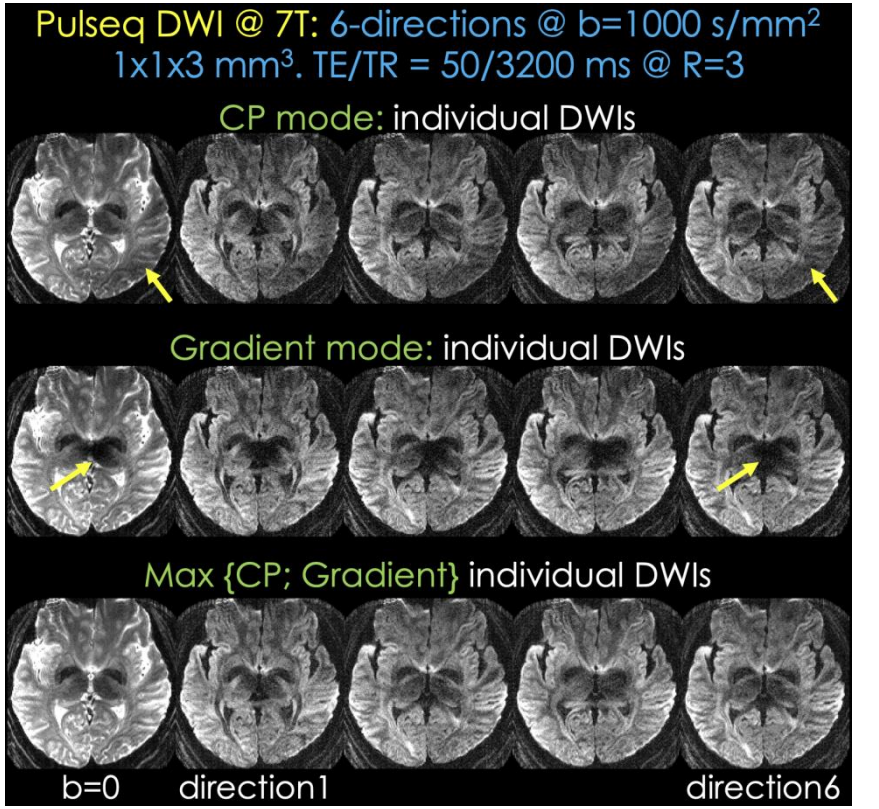
phase shift: 0 pi/2 pi -pi/2

RF Shimming at 7T

- In cooperation with Berkin Bilgic & Bastien Guerin @MGH



(Top) Individual TX channels selected manually for GRE
(Bottom) PulseqGRE with integrated shim vector control through the RF-Shim-Extension within a single sequence



In-vivo RF shimming with TIAMO

Summary of New Features

Readily work:

- Improved accuracy of the RF pulse information
- Improved gradient storage for non-Cartesian sequences
- More accurate and robust k-space calculation
- Rotation extension (especially) for non-Cartesian sequences
- Improved workflow by soft delay extension
- Dynamic RF shimming
- FOV positioning functionality in sequence design

In progress:

- More accurate FOV positioning in the (Siemens) interpreter (x-nuclei, non-Cartesian)

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Tony Stoecker
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Douglas Noll
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Lukas Winter
Naveen Murthy
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Sebastian Littin
Yogesh Rathi

THANK YOU FOR YOUR ATTENTION!

Pulseq MATLAB

pypulseq Python

harmonized MRI

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